PHOTON

Laser Tag Management System

TEAM 20 // SOFTWARE ENGINEERING

MEET THE TEAM

Team 20

Ryoji Araki	Structured repository and set up database interface logic.
Muhammad Ali	Designed UDP server and broadcasting logic.
Thomas Hamilton	GUI design & real-time display components.
Wahid Sultani	Created scripts necessary for project and managed libraries.

OUR APPROACH AS A TEAM

AGILE

- Trello
- Discord Chat
- SCRUM meetings as often as possible
- Customer feedback

What did we learn?

- Deadlines cannot be met all the time
- Problem decomposition is crucial
- Clear communication and unified goals are ingredients for cohesive software

THE PROJECT

What is Photon?

Photon is a real-time laser tag game management system designed for commercial laser tag arenas. It handles player registration, team setup, live scoring, and game countdowns which is all displayed on a high-visibility scoreboard screen visible to players and spectators alike.

- ✅ Live game timer
- ✅ Event feed
- ✅ Instant updates

GAME SETUP

Pre-Game: Team & Player Configuration

New Game

Start Game

RED TEAM

	Player ID	Codename	Equipment ID
Player #1	1	Viper	1
Player #2	2	Cobra	3
Player #3	3	Blaze	5
Player #4	4	Inferno	7
Player #5	5	Ember	9
Player #6	6	Scorch	11
Player #7	7	Crimson	13
Player #8	8	Fury	15
Player #9	9	Havoc	17
Player #10	10	Reaper	19
Player #11	11	Torch	21
Player #12	12	Flame	23
Player #13	13	Magma	25
Player #14	14	Cinder	27
Player #15	15	Ash	29

GREEN TEAM

	Player ID	Codename	Equipment ID
Player #1	16	Ghost	2
Player #2	17	Cipher	4
Player #3	18	Frost	6
Player #4	19	Storm	8
Player #5	20	Titan	10
Player #6	21	Echo	12
Player #7	22	Wraith	14
Player #8	23	Nova	16
Player #9	24	Specter	18
Player #10	25	Atlas	20
Player #11	26	Phantom	22
Player #12	27	Shadow	24
Player #13	28	Mirage	26
Player #14	29	Rogue	28
Player #15	30	Stealth	30

Red Team: up to 15 players

Green Team: up to 15 players

Player ID saves to database, allowing codename look up. Equipment ID can change from game to game.

Unique codenames for each player

New Game and Start Game controls and a responsive design that can be full-screen or shrunk down to a widget.

GAME LOBBY

Pre-Game Countdown & Scoreboard Initialization

30-second ready up phase

Red Team roster pre-loaded

Green Team roster pre-loaded

All scores initialized to zero

LIVE GAMEPLAY

Real-Time Score Updates & Event Feed

6-minute gameplay timer with 8 different Sci-Fi music tracks to play along to!

Live event feed (hits, friendlies)

Player score glows on point change, Winning team score flashes during game. “Game on!” and other phase messages pulse a glow around it.

Friendly fire results in a penalty of a 10-point deduction from hitter and receiver. Team net deduction is 20.

Scores sorted in real-time

SCORING IN ACTION

Green Team dominates the opening engagements

SCORING SYSTEM

Enemy Player Hit

Tag an opponent with your laser

+10

Enemy Base Hit

Score big by hitting the opposing base

+100

Friendly Fire

The teammate being hit and the teammate hitting receive this deduction

−10

TECH STACK

FRONTEND	**PyQt6 (Python GUI)** — Real-time scoreboard UI and player entry screen.
BACKEND	**Python** — Game logic, player management, scoring system, database interfacing
DATABASE	**PostgreSQL** — Each player gets their ID and Codename saved to a database for future sessions.
COMMS	UDP Sockets — Equipment ↔ Server hit registration
DEVOPS	Git / GitHub — Version control & collaboration
TESTING	Unit tests, integration tests, customer feedback

CHALLENGES & SOLUTIONS

⚠ Environment problems with install scripts

Fixed setup issues by using different install scripts for Mac and Windows/Linux. Corrected file formatting and made sure scripts run properly every time.

✓ SOLVED

⚠ Graphics Segmentation Faulting

Calling stop timer and delete timer at the same time caused a race condition; separated the call logic. Tracked timers cancel state using tuples of ID and Boolean cancelled variable. Added checks to prevent using deleted timers.

✓ SOLVED

⚠ Virtual Machine Woes...

We regularly communicated with team members, shared problems, asked for suggestions, and worked together to find solutions.

✓ SOLVED

⚠ General Debugging

We debugged step by step, tested each component separately, and added safety checks and logging to find and fix issues

✓ SOLVED

THANK YOU

Team 20 | Photon Laser Tag System

Questions?

github.com/SE-Team-20/photon-main

1 ~ 4: ryoji
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8~10: ali
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